

batata



RISC-V 32 BITS

Laboratório de sistemas computacionais:
Arquitetura e Organização de computadores

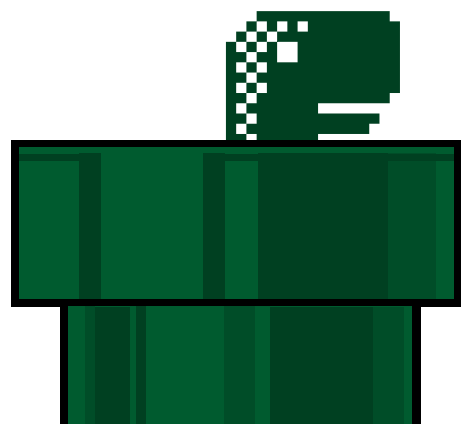
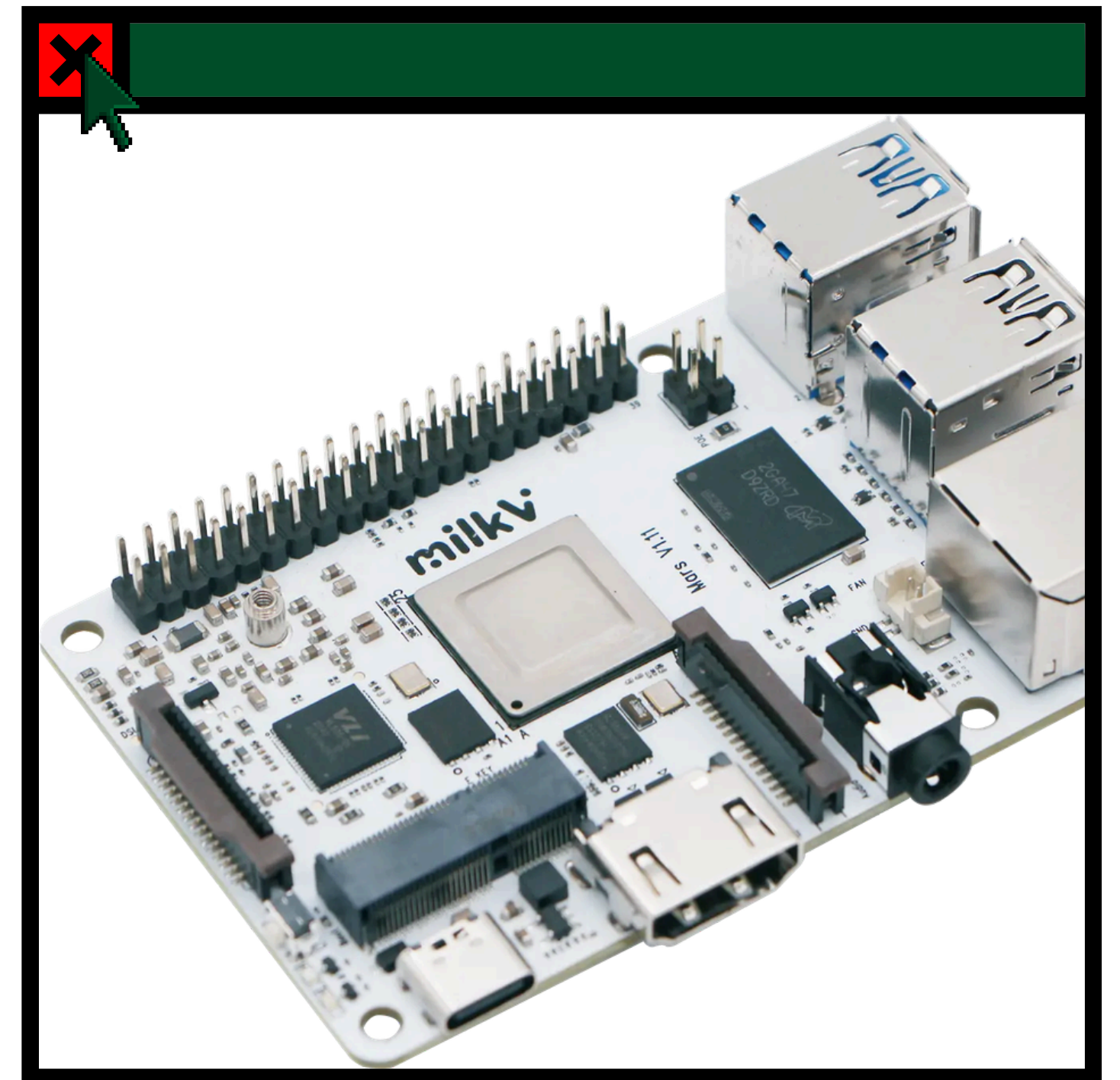
Leonardo Tomasela Leal
170291



pizza

SUMÁRIO

- Motivação
- Fundamentação teórica
- Arquitetura do computador
- Organização do computador



ORIGEM

Onde?

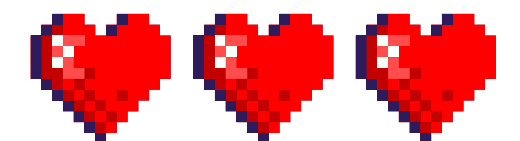
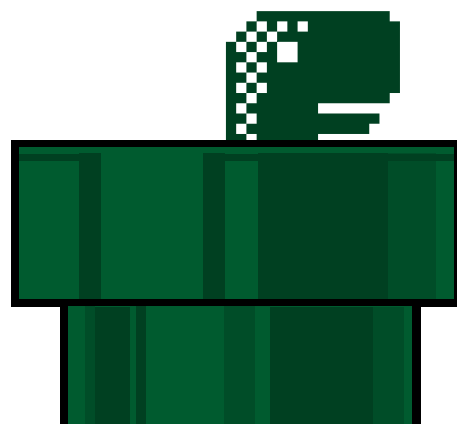
Foi criado na universidade da Califórnia em Berkeley (UC Berkeley), verão de 2010

Por que?

A equipe precisava de um processador moderno e de licença gratuita para usar em aulas e pesquisas

Por quem?

Projeto liderado por pesquisadores como Krste Asanović e David Patterson



POR QUE USAR RISC-V?

Gratuito

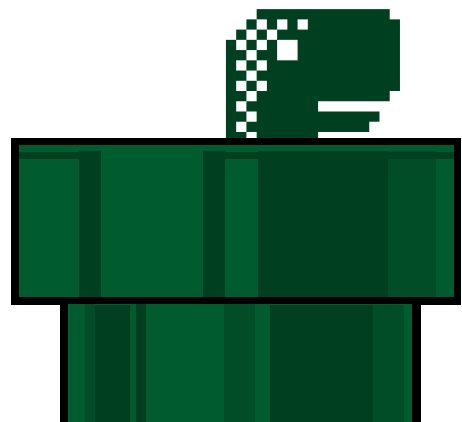
Como visto anteriormente, o principal motivo do uso do RISC-V é ele ser uma alternativa open source a arquiteturas competidoras, como x86 e ARM.

Modular

O RISC-V é um conjunto de instruções modular, com várias extensões, o que o torna fácil de adaptar para diversos usos.

Estilo de vida

É open source.

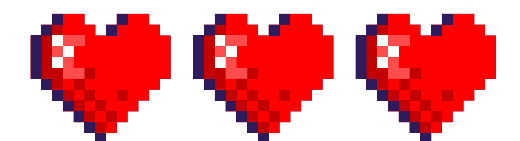
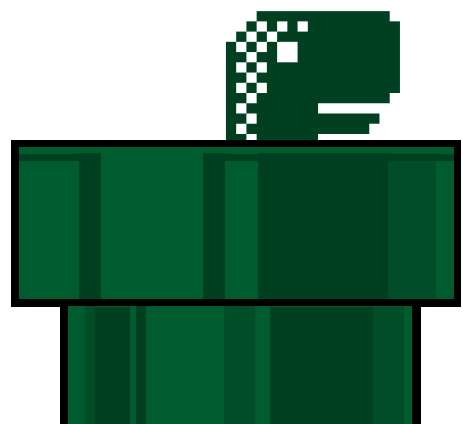
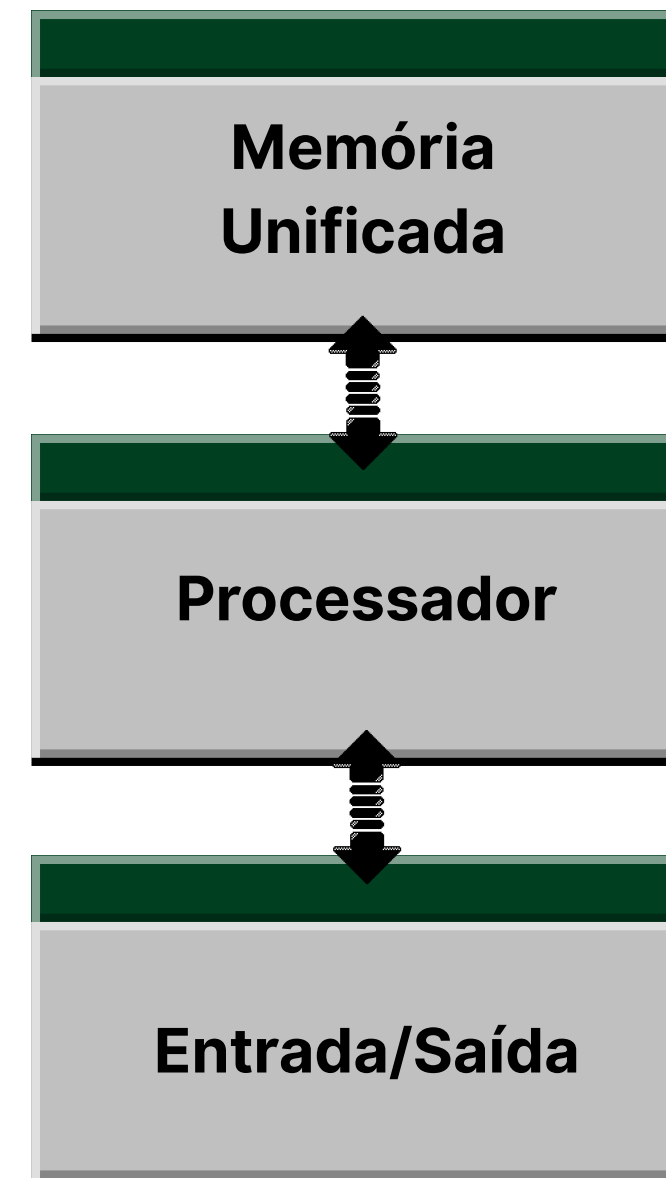


FUNDAMENTAÇÃO TEÓRICA

Arquitetura de harvard x Arquitetura de von neumann

Arquitetura de von neumann:

- Memória de dados e memória de instruções unificada
- Barramento único para ambos
- Execução das instruções em ordem sequencial

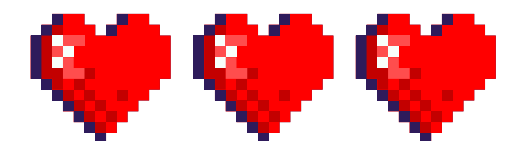
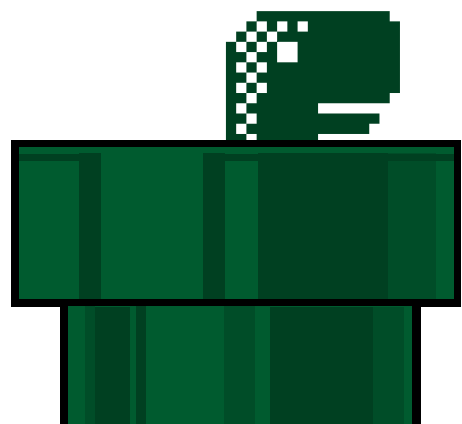
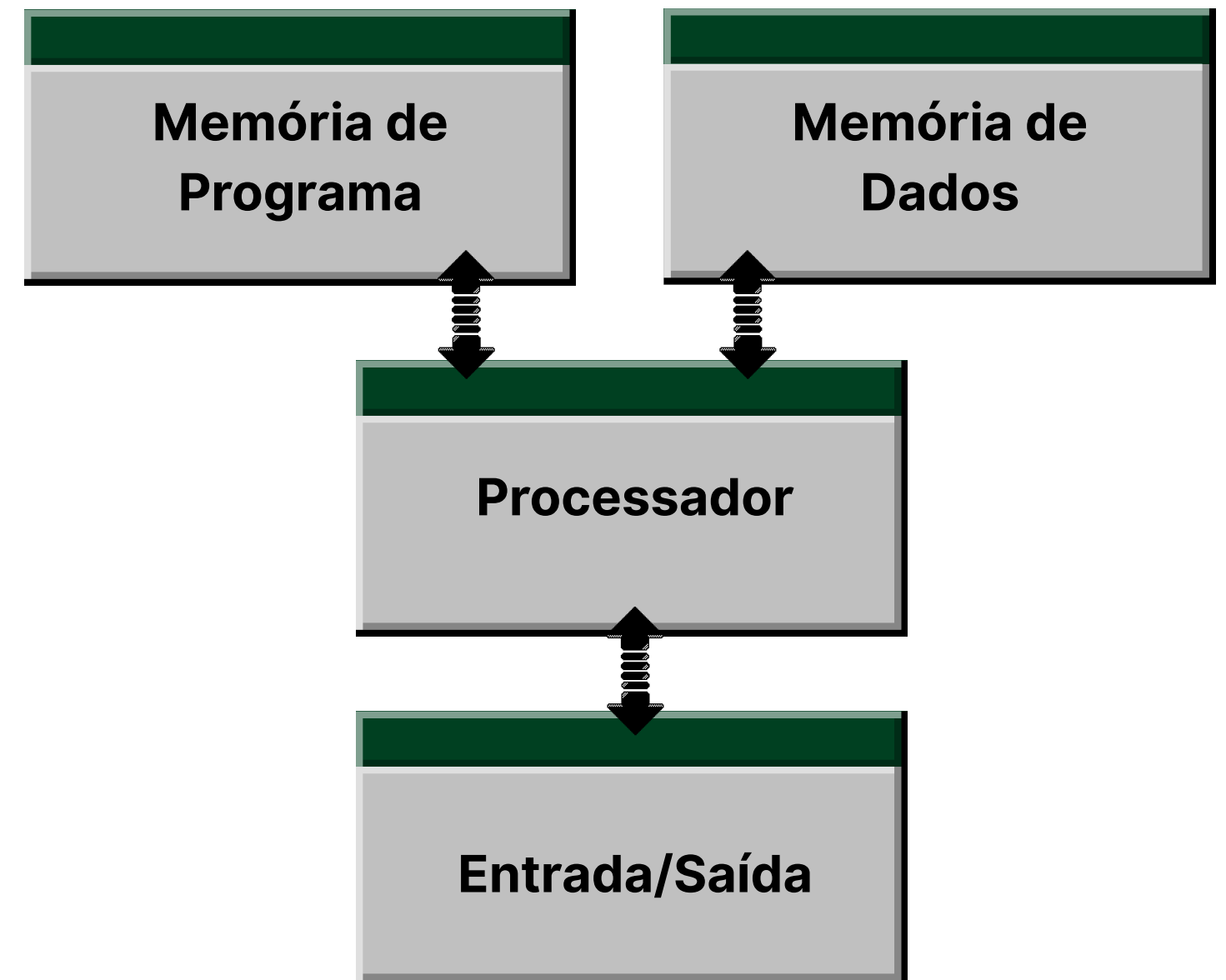


FUNDAMENTAÇÃO TEÓRICA

Arquitetura de harvard x Arquitetura de von neumann

Arquitetura de harvard:

- Memória de dados e memória de instruções separada
- Barramentos separados, possibilitando busca de instruções e dados ao mesmo tempo



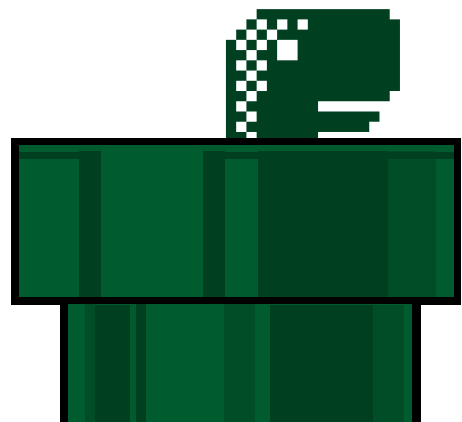
RISC VS CISC

RISC

- Reduced instruction set computer
- Arquitetura load store
- Arquitetura de harvard
- Tamanho fixo das instruções
- Poucas instruções mais simples
- Poucos modos de endereçamento
- Vários registradores

CISC

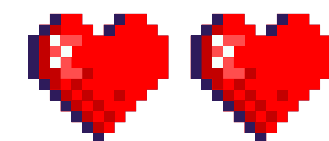
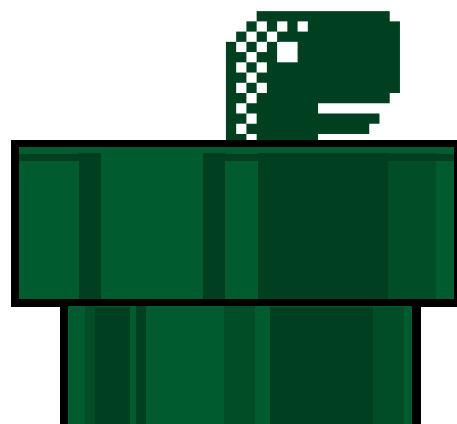
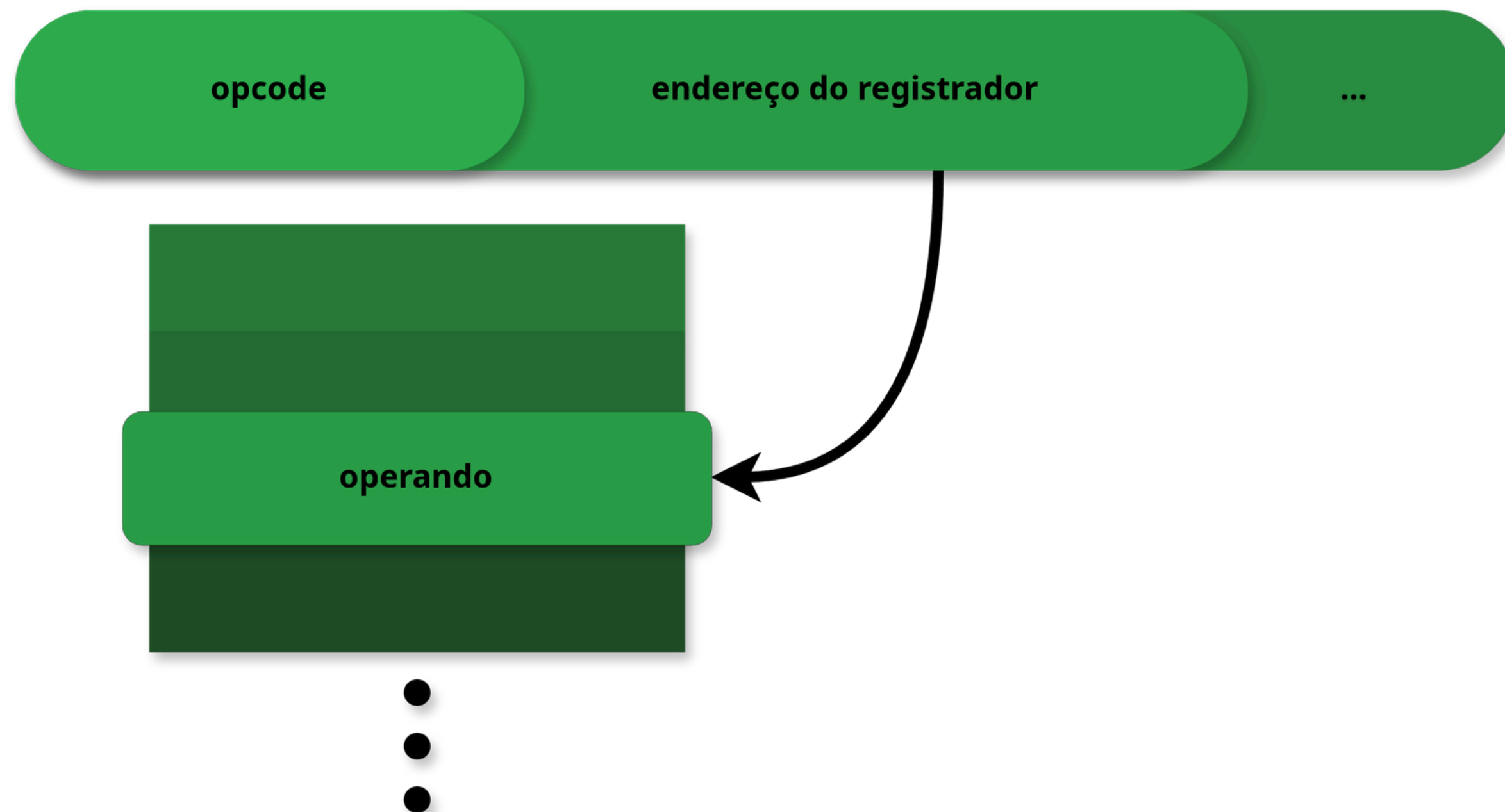
- Complex instruction set computer
- Arquitetura de von neumann
- Tamanho variado das instruções
- Muitas instruções complexas
- Vários modos de endereçamento
- Poucos registradores



MODOS DE ENDEREÇAMENTO

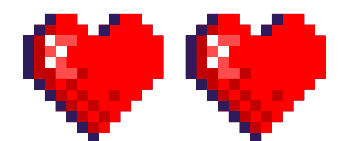
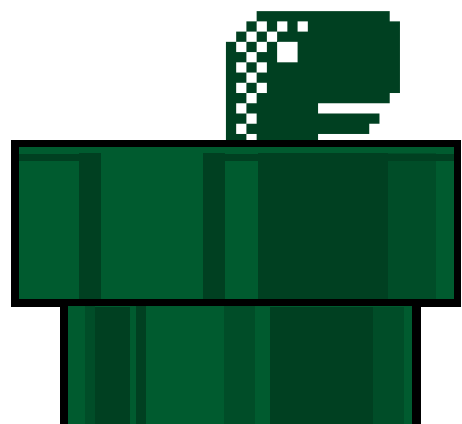
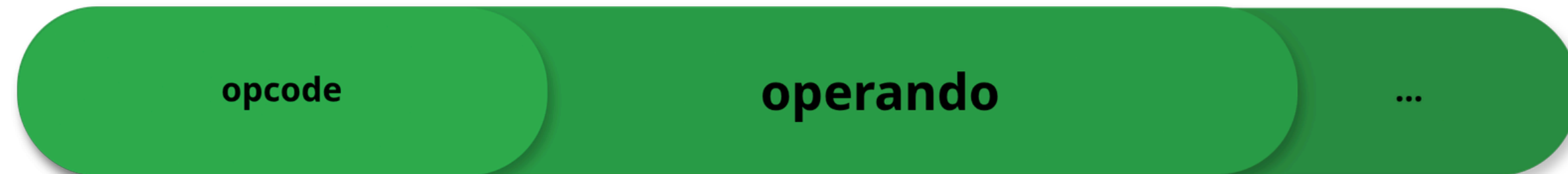
Endereçamento por registrador

- A instrução referencia o registrador, que contém as instruções.



MODOS DE ENDEREÇAMENTO

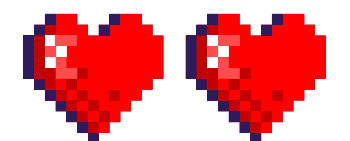
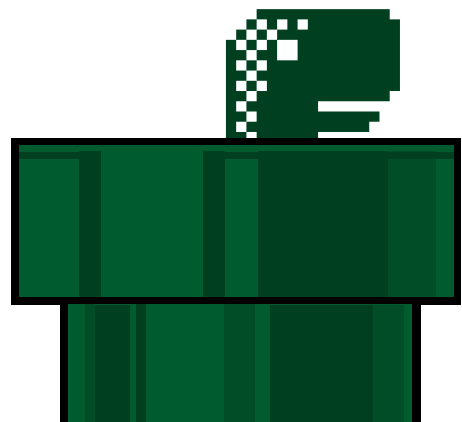
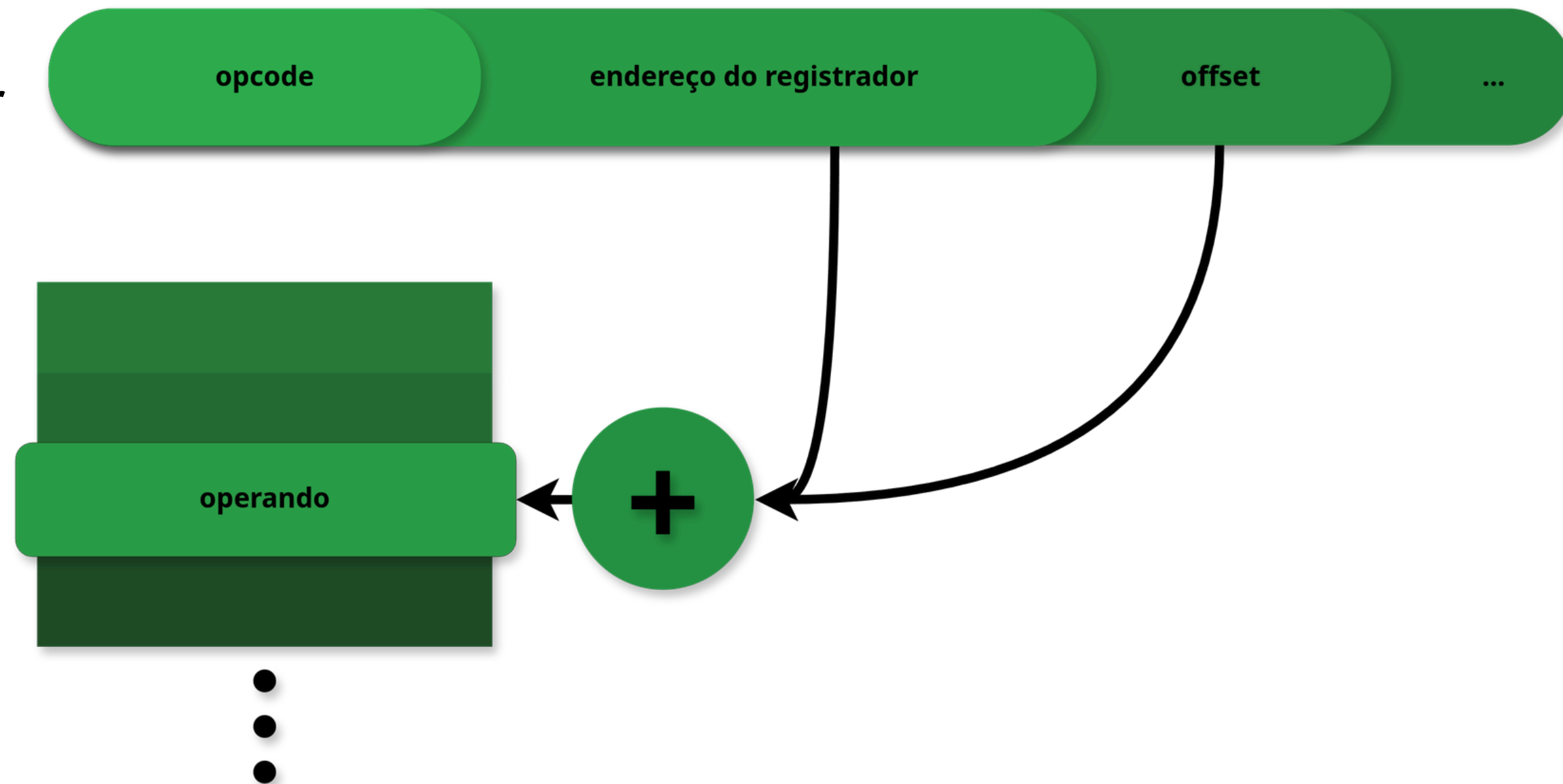
-  **Endereçamento por imediato**
- O valor do operando é contido nas instruções.



MODOS DE ENDEREÇAMENTO

Endereçamento por deslocamento

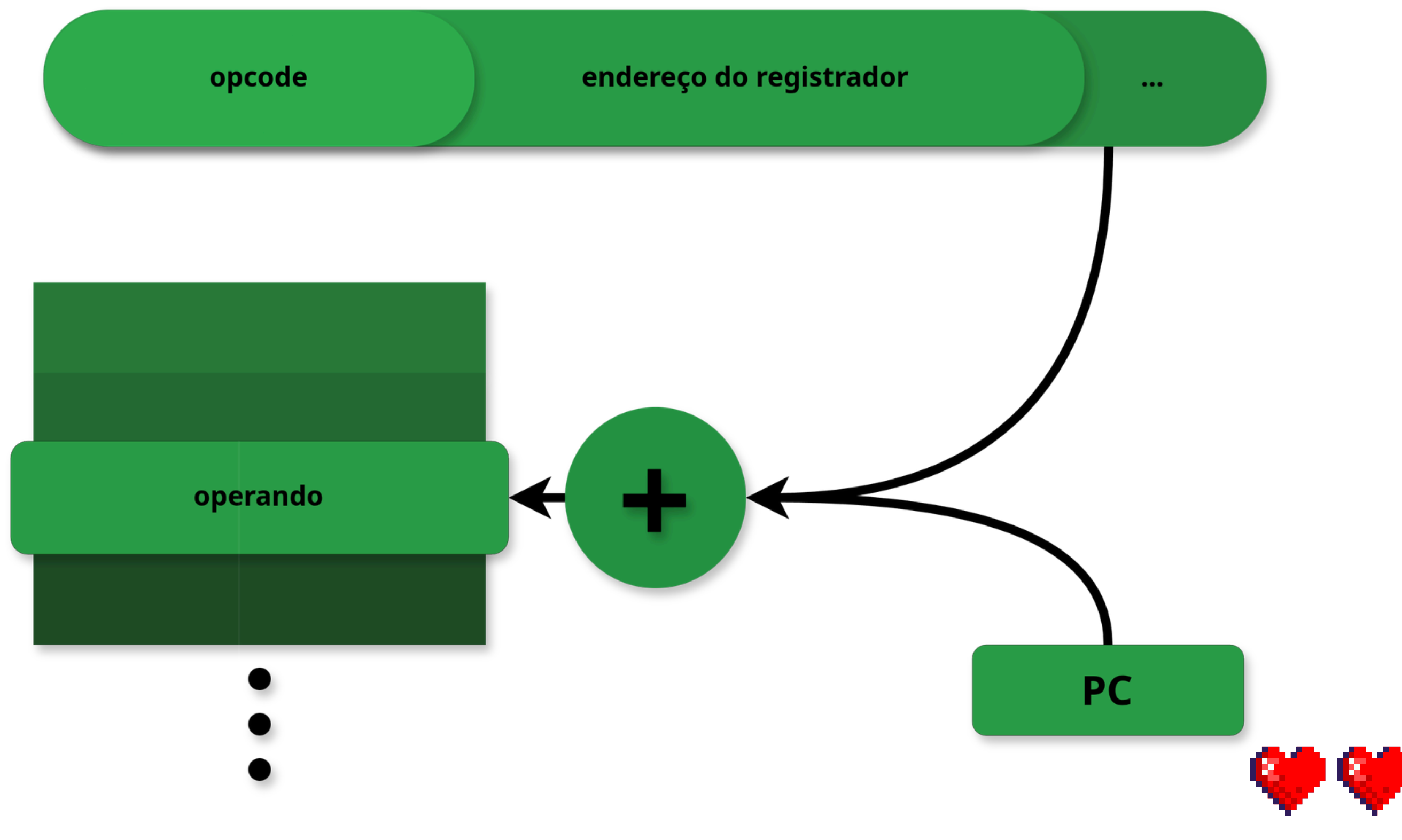
- A memória é acessada somando um valor base contido em um registrador a um imediato de 12 bits.



MODOS DE ENDEREÇAMENTO

Endereçamento pc-relativo

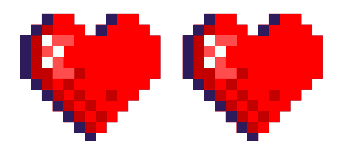
- O endereço de destino é obtido somando o valor contido no pc a um imediato.



abacaxi do ru

ARQUITETURA RISC-V

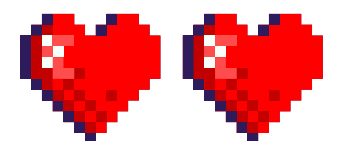
-  Arquitetura RISC
-  Design Modular
-  Possui padrão Aberto (Open Source)
-  Possui 32 registradores de uso geral



peixe

ARQUITETURA RISC-V

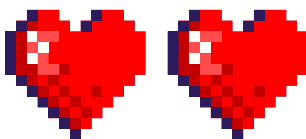
-  Instruções de 32 bits
-  Formato simples de instruções
-  Endereçamento por byte
-  little endian



ARQUITETURA RISC-V

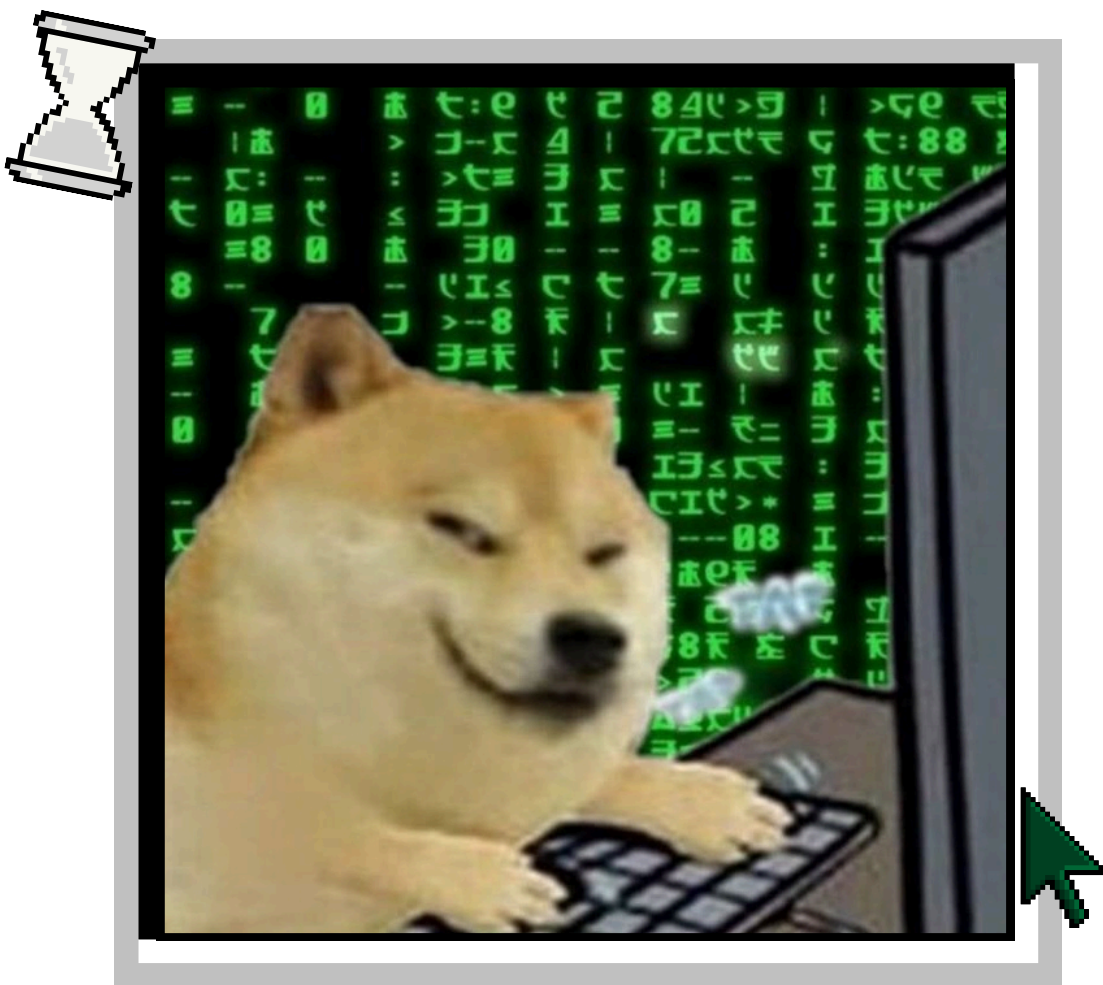
Formato de instruções:

31	25	24	20	19	15	14	12	11	7	6	0	
funct7		rs2		rs1		funct3		rd		opcode		Tipo R
imm[11:0]				rs1		funct3		rd		opcode		Tipo I
imm[11:5]		rs2		rs1		funct3		imm[4:0]		opcode		Tipo S
imm[11:5]		rs2		rs1		funct3		imm[4:0]		opcode		Tipo B
imm[31:12]										opcode		Tipo U

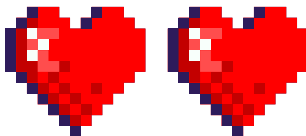


alçafrão

INSTRUÇÕES DO TIPO R

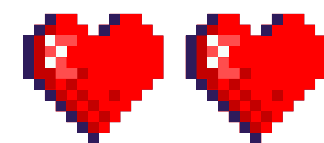


Mnemônico	funct3	funct7	Opcode	Descrição	Notas
add	0x0	0x00	0110011	$rd = rs1 + rs2$	
sub	0x0	0x20	0110011	$rd = rs1 - rs2$	
xor	0x4	0x00	0110011	$rd = rs1 \wedge rs2$	
or	0x6	0x00	0110011	$rd = rs1 \text{ --- } rs2$	
and	0x7	0x00	0110011	$rd = rs1 \& rs2$	
sll	0x1	0x00	0110011	$rd = rs1 \ll rs2$	
srl	0x5	0x00	0110011	$rd = rs1 \gg rs2$	
sra	0x5	0x20	0110011	$rd = rs1 \gg rs2$	msb-extends
slt	0x2	0x00	0110011	$rd = (rs1 < rs2)?1:0$	
sltu	0x3	0x00	0110011	$rd = (rs1 < rs2)?1:0$	zero-extends
mul	0x0	0x01	0110011	$rd = (rs1 * rs2)[31:0]$	
mulh	0x1	0x01	0110011	$rd = (rs1 * rs2)[63:32]$	
mulsu	0x2	0x01	0110011	$rd = (rs1 * rs2)[63:32]$	
mulu	0x3	0x01	0110011	$rd = (rs1 * rs2)[63:32]$	
div	0x4	0x01	0110011	$rd = rs1 / rs2$	
divu	0x5	0x01	0110011	$rd = rs1 / rs2$	
rem	0x6	0x01	0110011	$rd = rs1 \% rs2$	
remu	0x7	0x01	0110011	$rd = rs1 \% rs2$	



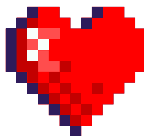
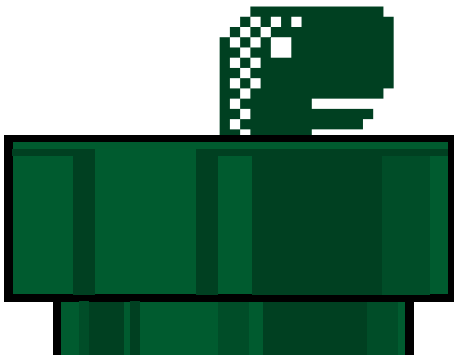
INSTRUÇÕES DO TIPO I

Mnemônico	funct3	imm	Opcode	Descrição	Notas
addi	0x0		0010011	$rd = rs1 + imm$	
xori	0x4		0010011	$rd = rs1 \wedge imm$	
ori	0x6		0010011	$rd = rs1 \mid imm$	
andi	0x7		0010011	$rd = rs1 \& imm$	
slli	0x1	$imm[5:11]=0x00$	0010011	$rd = rs1 \ll imm[0:4]$	
srli	0x5	$imm[5:11]=0x00$	0010011	$rd = rs1 \gg imm[0:4]$	
srai	0x5	$imm[5:11]=0x20$	0010011	$rd = rs1 \ggg imm[0:4]$	msb-extends
slti	0x2		0010011	$rd = (rs1 < imm)?1:0$	
sltiu	0x3		0010011	$rd = (rs1 < imm)?1:0$	zero-extends
lb	0x0		0000011	$rd = M[rs1+imm][0:7]$	
lh	0x1		0000011	$rd = M[rs1+imm][0:15]$	
lw	0x2		0000011	$rd = M[rs1+imm][0:31]$	
lbu	0x4		0000011	$rd = M[rs1+imm][0:7]$	zero-extends
lhu	0x5		0000011	$rd = M[rs1+imm][0:15]$	zero-extends
jalr	0x0		1100111	$rd = PC+4; PC = rs1 + imm$	



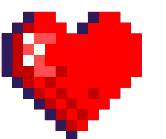
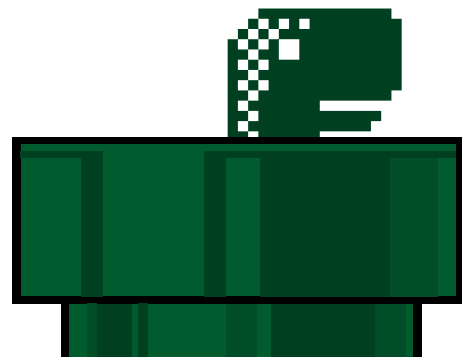
INSTRUÇÕES DO TIPO S

Mnemônico	funct3	Opcode	Descrição
sb	0x0	0100011	$M[rs1+imm][0:7] = rs2[0:7]$
sh	0x1	0100011	$M[rs1+imm][0:15] = rs2[0:15]$
sw	0x2	0100011	$M[rs1+imm][0:31] = rs2[0:31]$



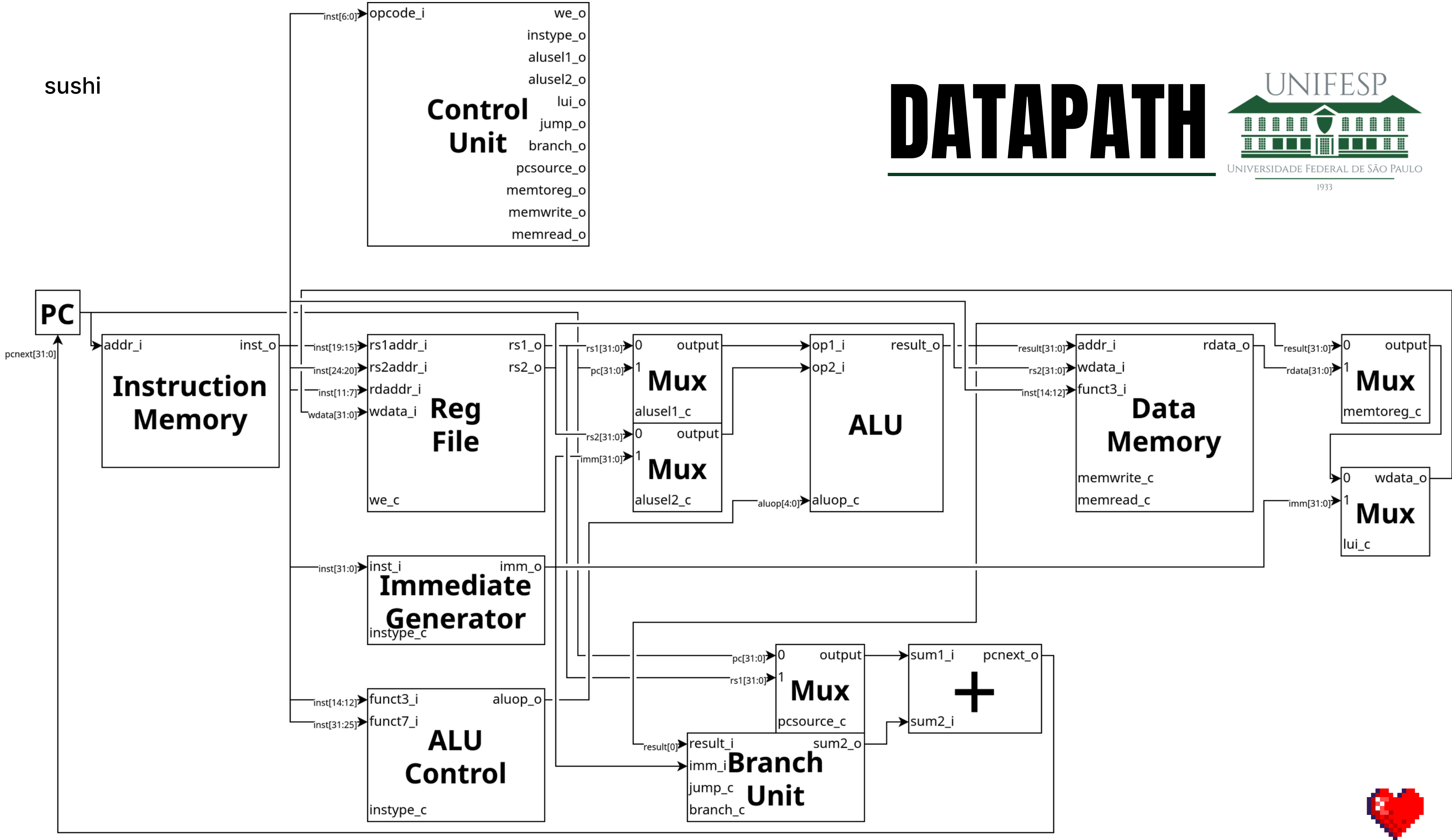
INSTRUÇÕES DO TIPO U

Mnemônico	Opcode	Descrição
lui	0110111	$rd = imm \ll 12$
auipc	0010111	$rd = PC + (imm \ll 12)$
jal	1101111	$rd = PC + 4; PC += imm$



sushi

DATAPATH

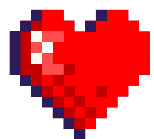
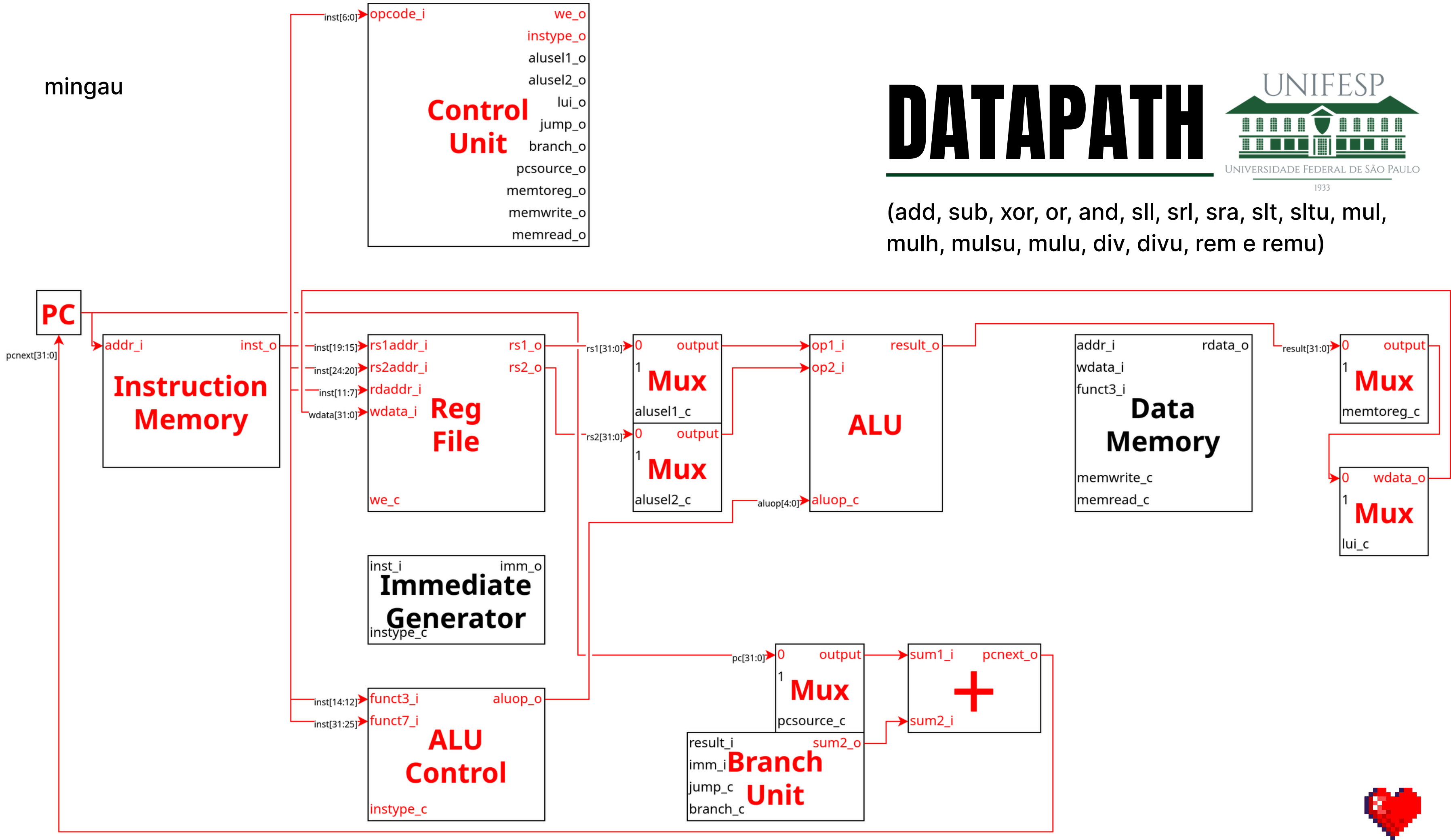


mingau

DATAPATH



(add, sub, xor, or, and, sll, srl, sra, slt, sltu, mul, mulh, mulsu, mulu, div, divu, rem e remu)

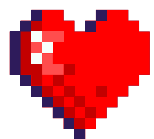
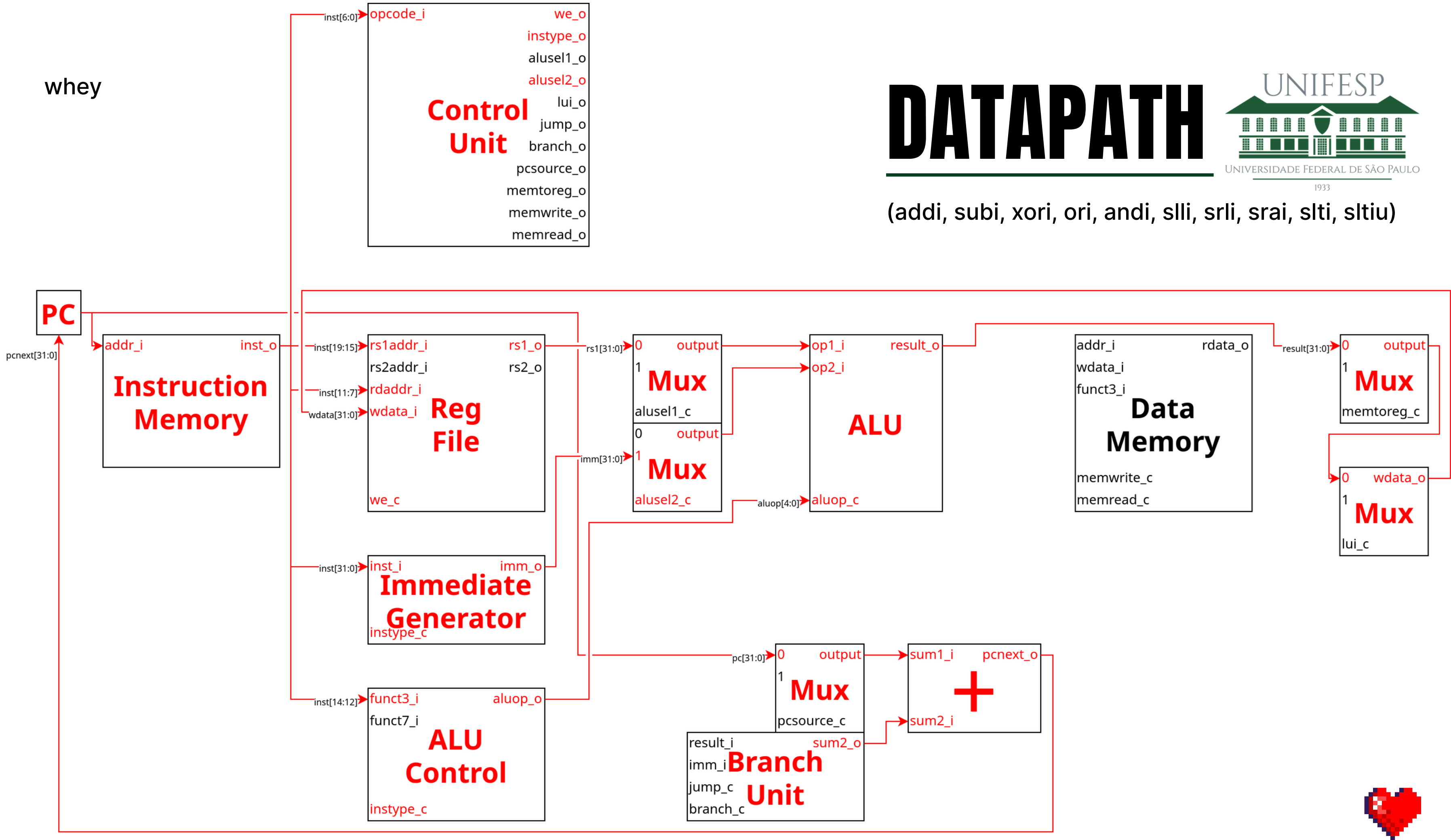


whey

DATAPATH



(addi, subi, xori, ori, andi, slli, srli, srai, slti, sltiu)

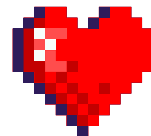
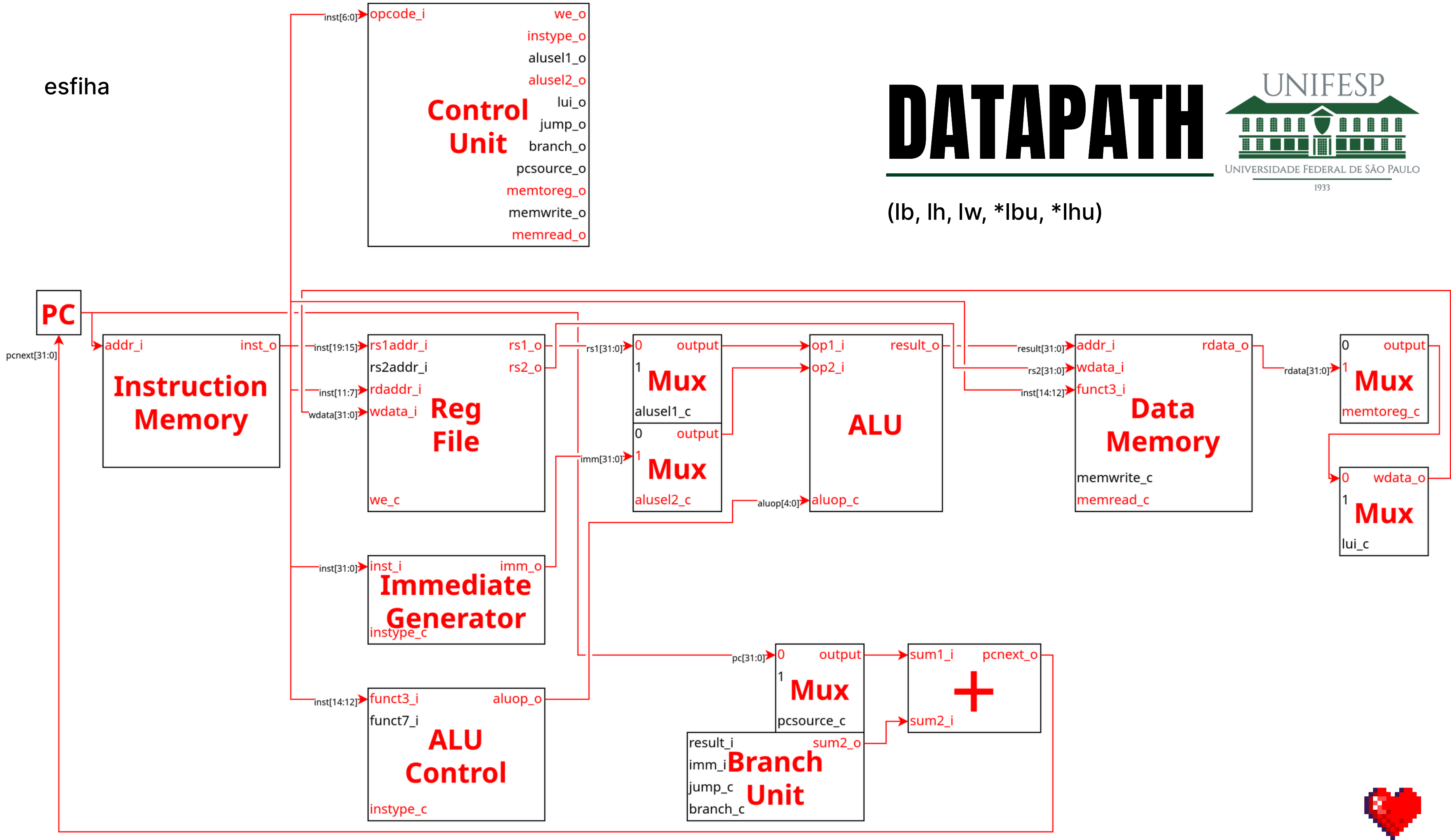


esfiha

DATAPATH



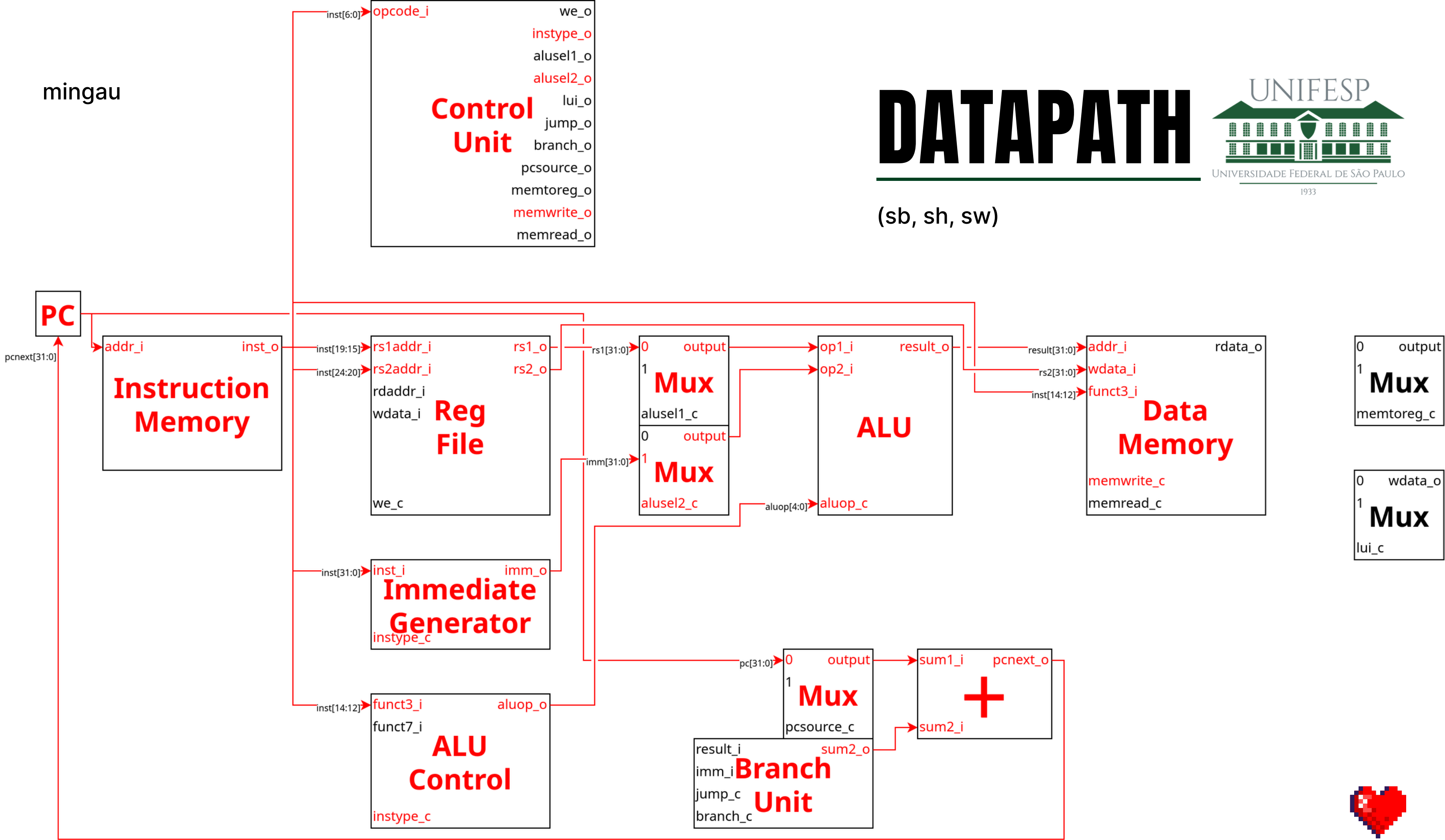
(lb, lh, lw, *lbu, *lhu)



mingau

DATAPATH

(sb, sh, sw)

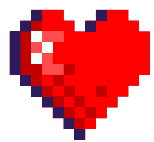
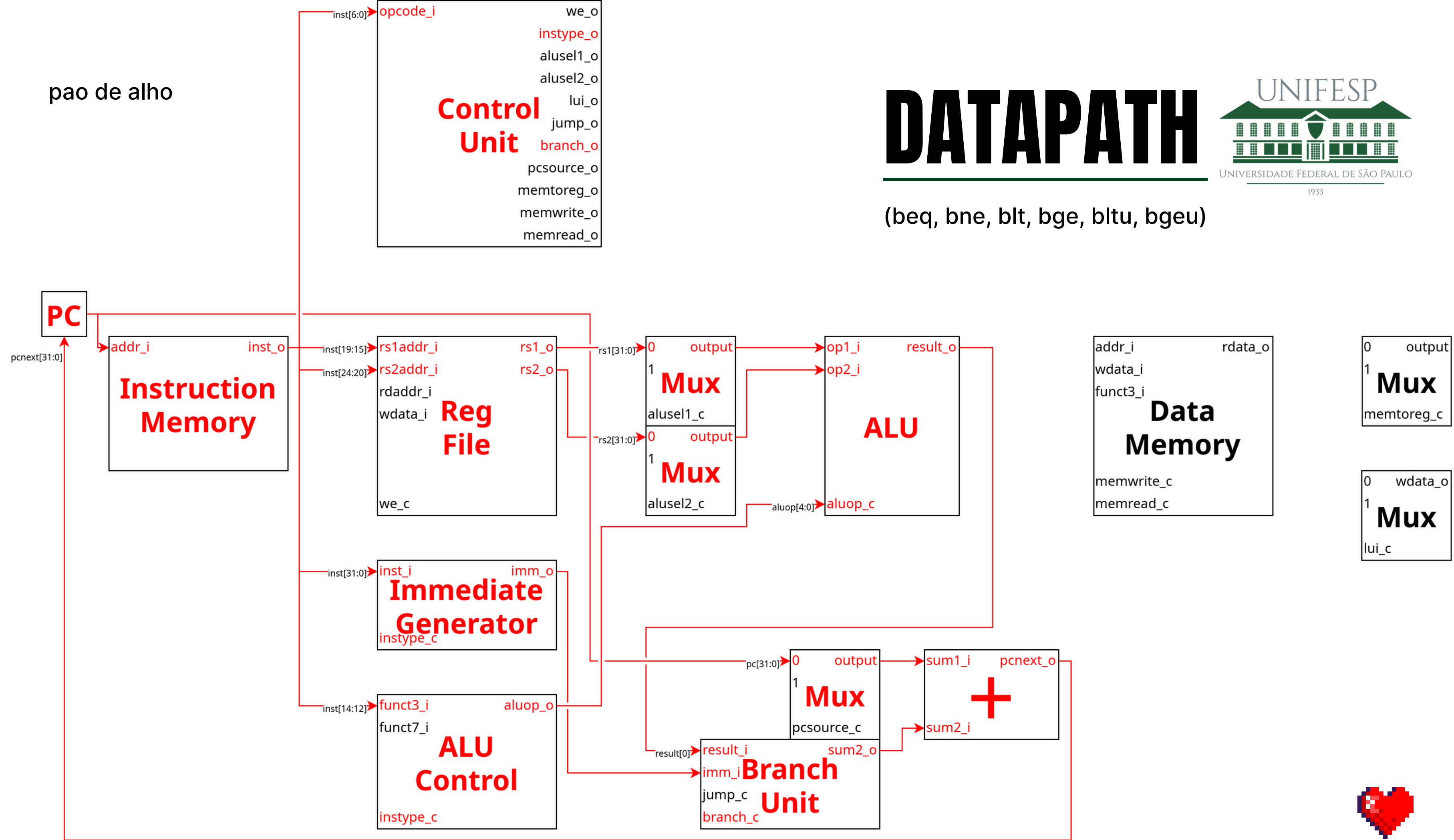


pao de alho

DATAPATH



(beq, bne, blt, bge, bltu, bgeu)

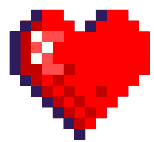
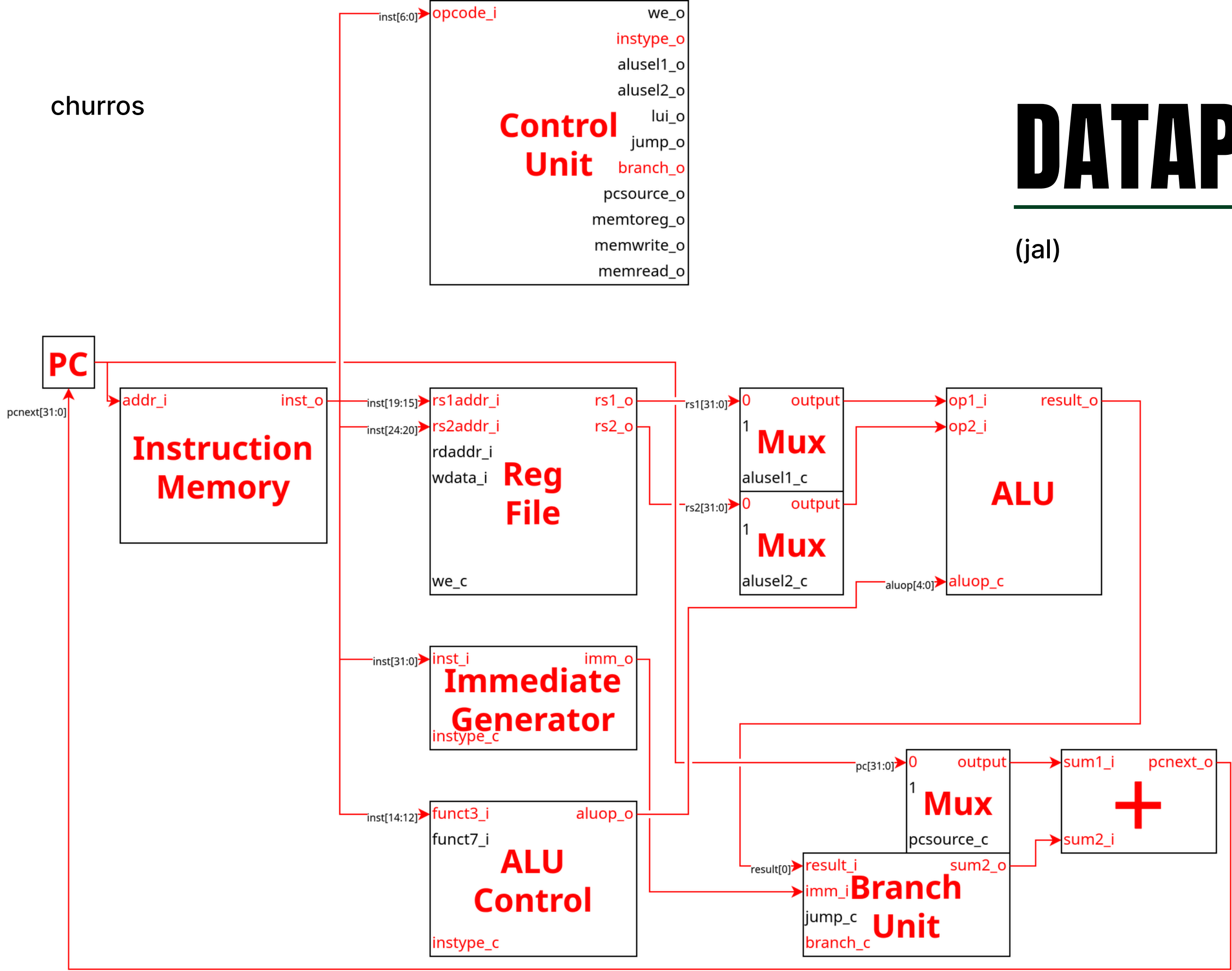


churros

DATAPATH



(jal)

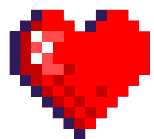
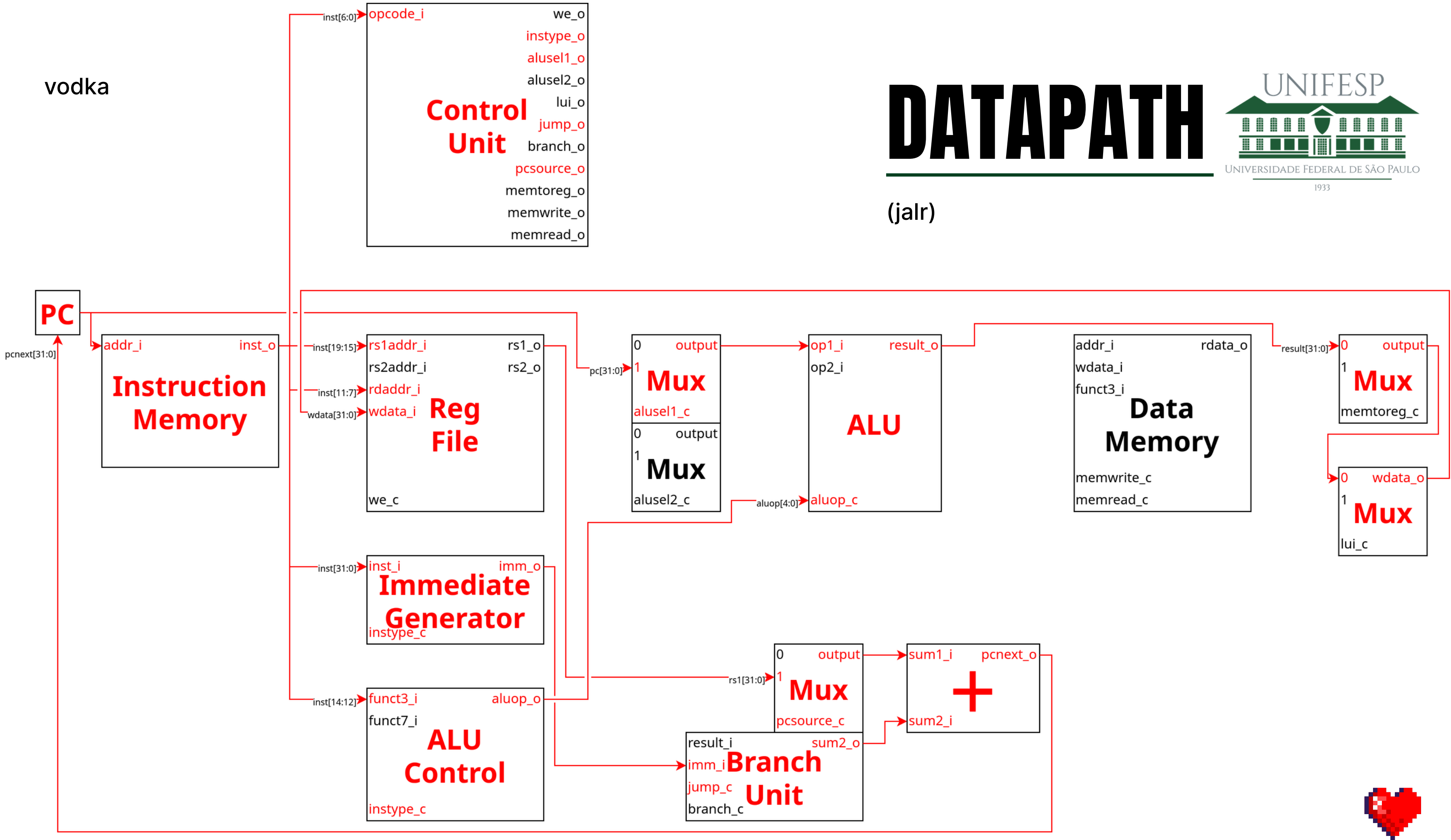


vodka

DATAPATH



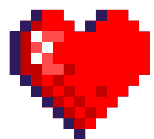
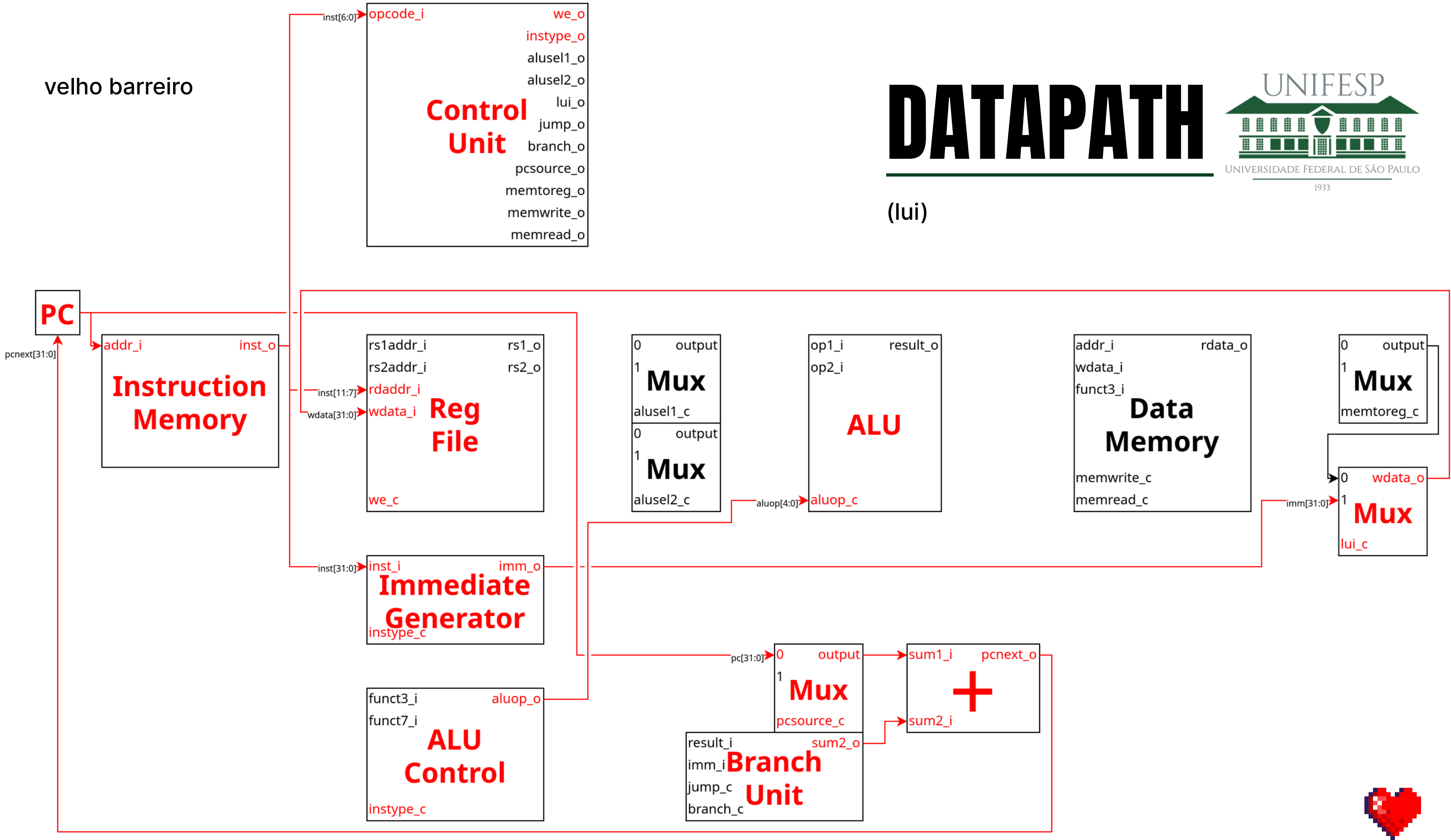
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velho barreiro

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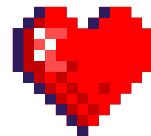
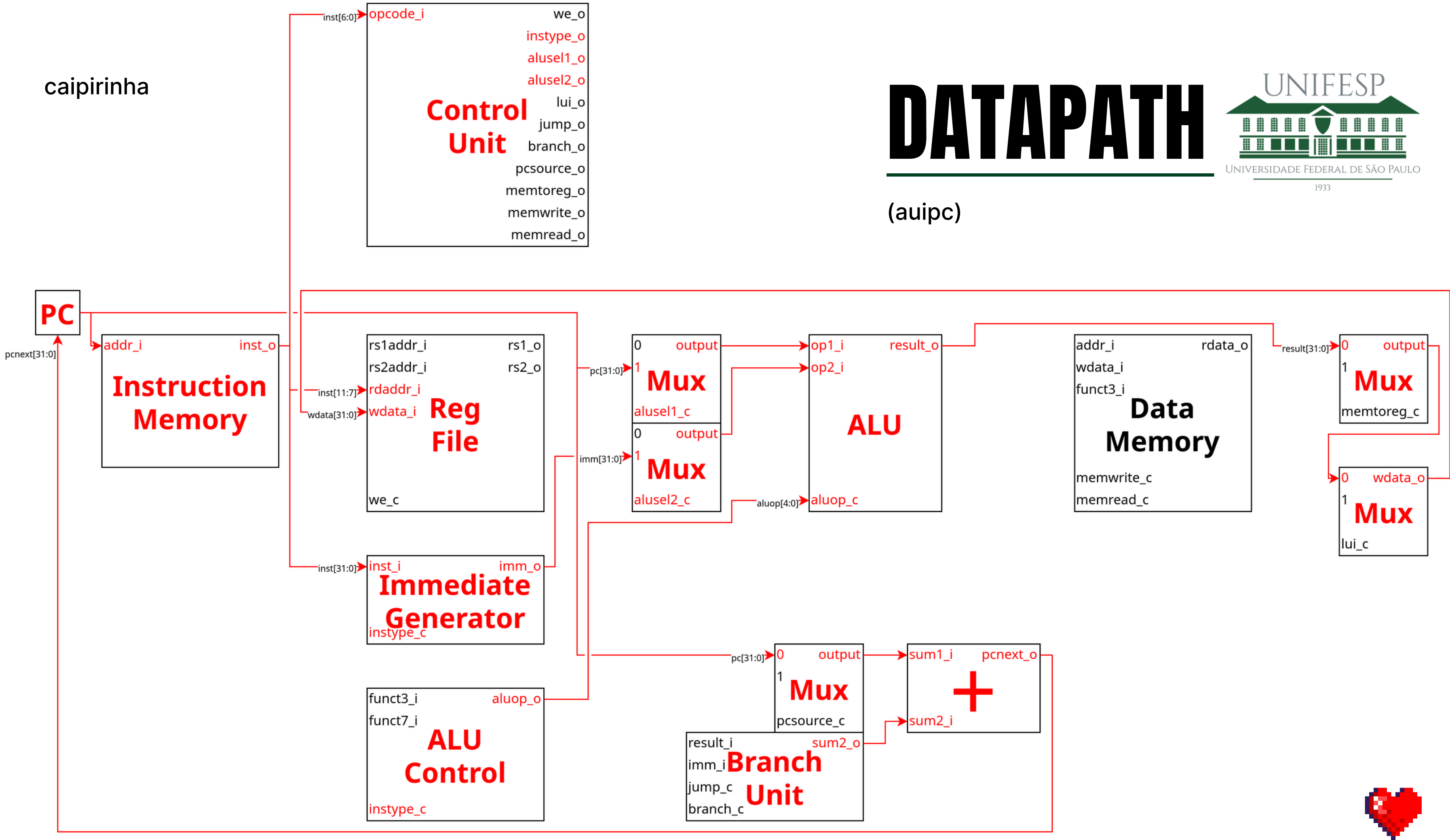
(lui)



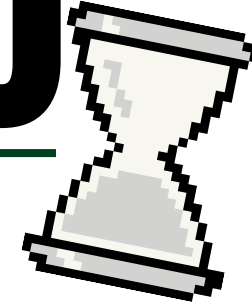
caipirinha

DATAPATH

(auipc)



OBRIGADO PELA ATENÇÃO



Fontes e referências

- **Flowchart Maker & Online Diagram Software.** Disponível em: <https://app.diagrams.net/#>. Acesso em: 1 abr. 2026.
- **LaTeX to Image converter.** Disponível em: <https://latex2image.joeraut.com/>. Acesso em: 1 abr. 2026.
- **RISC-V Foundation.** Disponível em: <https://riscv.org/>. Acesso em: 1 abr. 2026.
- **RISC-V Reference RISC-V Instruction Set.** Disponível em: https://www.cs.sfu.ca/~ashriram/Courses/CS295/assets/notebooks/RISCV/RISCV_CARD.pdf. Acesso em: 1 abr. 2026.
- **RISC-V.** Disponível em: <https://en.wikipedia.org/wiki/RISC-V>. Acesso em: 1 abr. 2026.
- **Five EmbedDev.** Disponível em: <https://five-embeddev.com/>. Acesso em: 1 abr. 2026.
- PATTERSON, D. A.; HENNESSY, J. L. **Computer organization and design RISC-V edition : the hardware software interface.** Amsterdam: Morgan Kaufmann, 2021.

